

ILOs

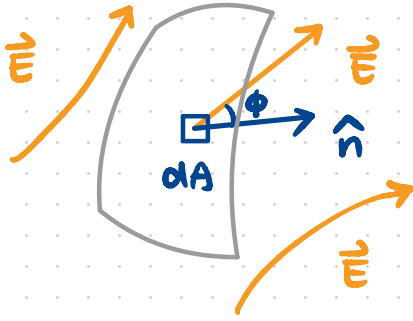
1. Apply Gauss's Law to calculate the electric field for charge distributions with spherical, cylindrical, and planar symmetry.
2. Apply Gauss's Law to determine the electric properties of conductors in electrostatic equilibrium.

Ch22 Gauss's Law

→ A fundamental statement about the relationship between electric charges and electric fields.

Electric Flux Φ_E

open surface

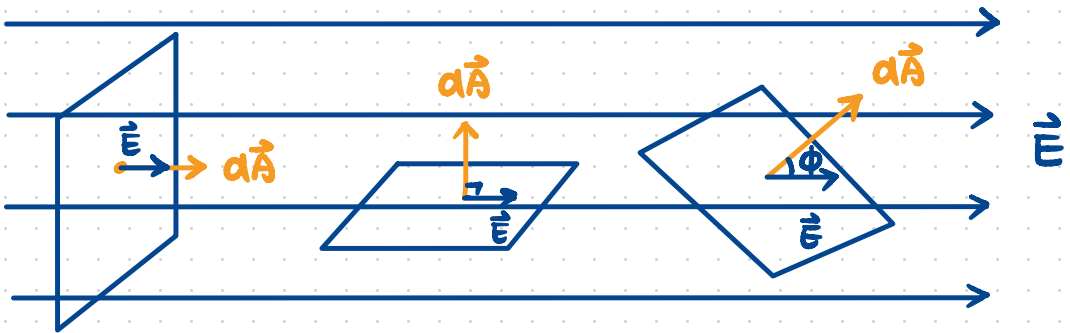


$$d\Phi_E \equiv \vec{E} \cdot \hat{n} dA = \vec{E} \cdot d\vec{A} \\ = E dA \cos\phi$$

$$\Phi_E = \int d\Phi_E \\ = \int \vec{E} \cdot d\vec{A}$$

$$\text{SI unit: } \frac{\text{N}}{\text{C}} \cdot \text{m}^2$$

Example: uniform $\vec{E}$, flat surface



$$\phi = 0$$

$$\Phi_E = EA$$

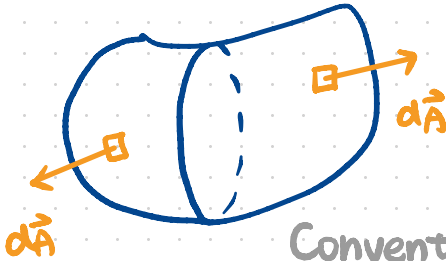
$$\phi = \frac{\pi}{2} (90^\circ)$$

$$\Phi_E = EA \cos\left(\frac{\pi}{2}\right) \\ = 0$$

$$\phi = \frac{\pi}{3} (60^\circ)$$

$$\Phi_E = EA \cos\left(\frac{\pi}{3}\right) \\ = \frac{1}{2} EA$$

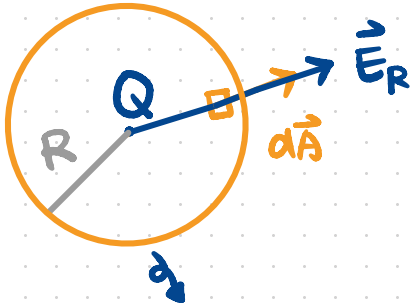
closed surface



$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

Convention: $d\vec{A}$ points outward

Example : Point Charge



closed surface

$$\Phi_E = \oint \vec{E}_R \cdot d\vec{A}$$

$$\text{Where } \vec{E}_R = \frac{Q}{4\pi\epsilon_0 R^2} \hat{r}$$
$$d\vec{A} = dA \hat{r}$$

$$= \oint E_R dA = E_R \oint dA$$

$$= 4\pi R^2 E_R$$

$$= \cancel{4\pi R^2} \frac{Q}{\cancel{4\pi\epsilon_0 R^2}}$$

$$\Rightarrow \frac{Q}{\epsilon_0} \begin{matrix} > \\ = \\ < \end{matrix} 0$$

Independent of R

$\therefore$ same number of electric field lines

Gauss's Law

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

ES Coulomb's Law

$$\equiv d\vec{E} = \frac{dq}{4\pi\epsilon_0 r^2} \hat{r}$$

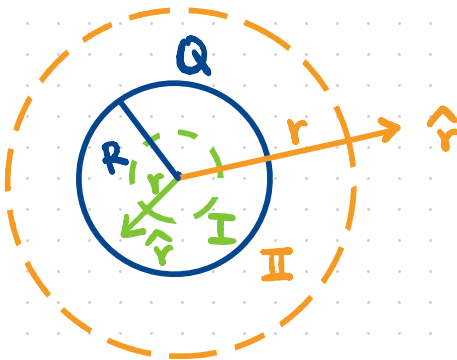
closed surface: Gaussian Surface

Symmetries

- (1) 球 spherical symmetry
- (2) 柱 cylindrical symmetry
- (3) 面 planar symmetry

Examples of (1) :

Uniform Spherical Shell



(I) $r < R$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$E \times 4\pi r^2 = 0$$

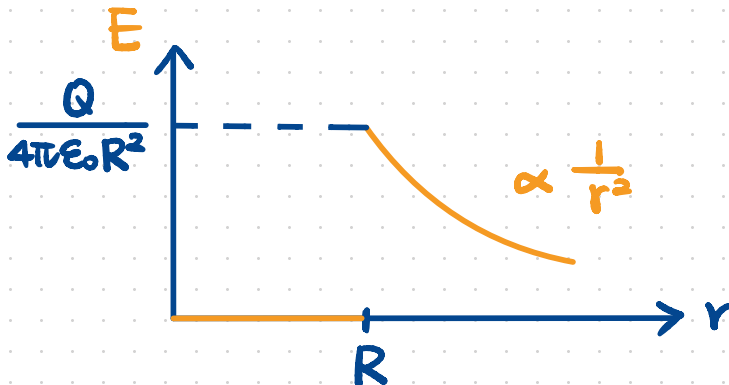
$$\Rightarrow \boxed{\vec{E} = \vec{0}} \quad (\text{Inside shell})$$

(II) $r > R$

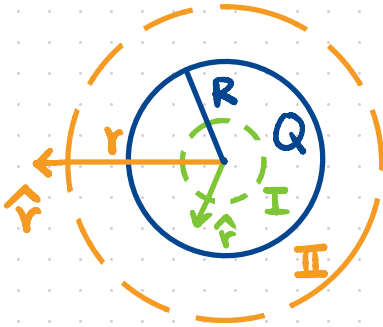
$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} \Rightarrow \boxed{\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}} \quad (\text{outside shell})$$

A shell of uniform charge attracts or repels a charged particle that is outside the shell as if all the shell's charge were concentrated at the center of the shell.



Uniform Sphere



volume charge density

$$\rho \equiv \frac{dq}{dv} = \frac{Q}{V} = \text{const}$$

↑
uniform

$$\Rightarrow \rho \times \frac{4}{3}\pi R^3 = Q$$

$$\Rightarrow \rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

(I) $r < R$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

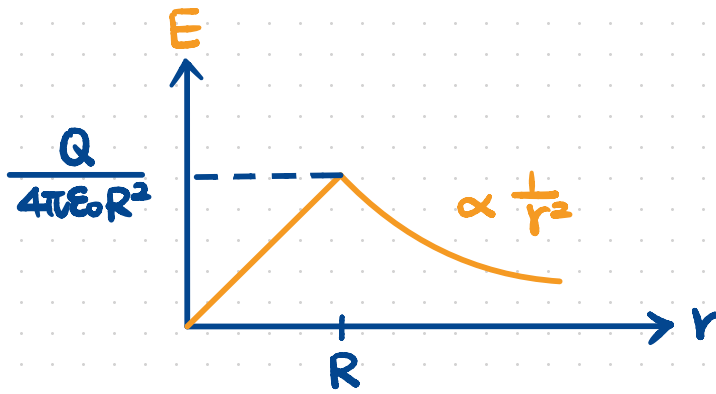
$$E \times \cancel{4\pi r^2} = \frac{\rho \times \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho r}{3\epsilon_0} = \frac{Q}{4\pi\epsilon_0 R^3} r \propto r$$

(II) $r > R$

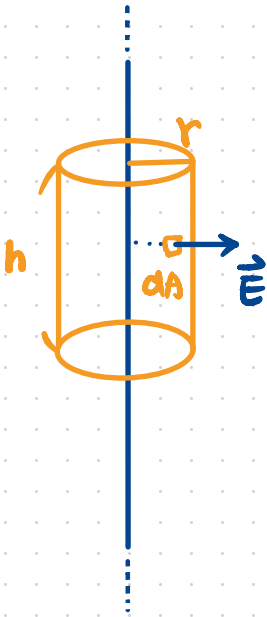
$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} \Rightarrow \vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$



Example of (2) : Infinite Long Uniform Line Charge

line charge density $\lambda = \text{const (C/m)}$



$$\begin{aligned}\Phi_E &= \Phi_t + \Phi_b + \Phi_{\text{curved}} \\ &= 0 + 0 + E \times 2\pi r h\end{aligned}$$

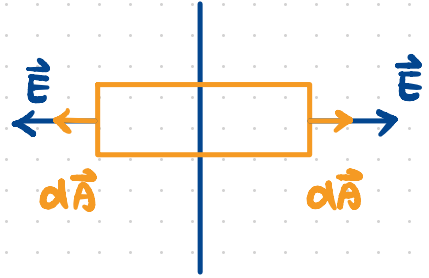
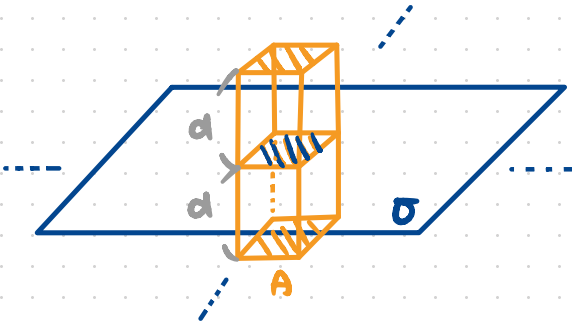
$$Q_{\text{encl}} = \lambda h$$

$$\Phi_E = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\Rightarrow E \times 2\pi r h = \frac{\lambda h}{\epsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r} \Rightarrow \vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} \propto \frac{1}{r}$$

Example of (3) : Infinite Uniform Sheet



surface charge density $\sigma = \text{const}$ (C/m^2)

$$\Phi_E = EA + EA = 2EA$$

$$Q_{\text{encl}} = \sigma A$$

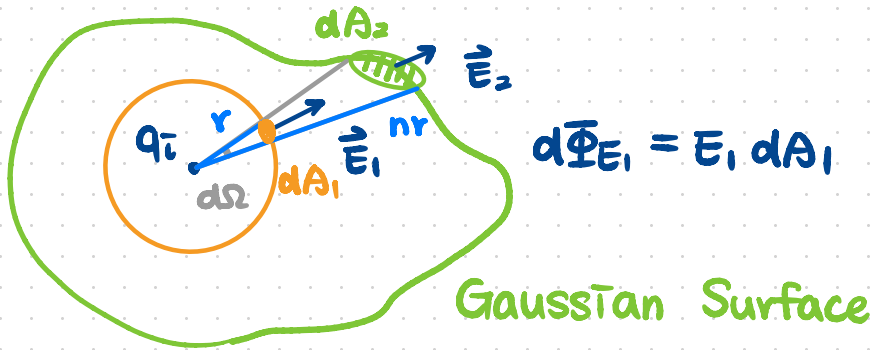
$$\Phi_E = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\Rightarrow 2EA = \frac{\sigma A}{\epsilon_0}$$

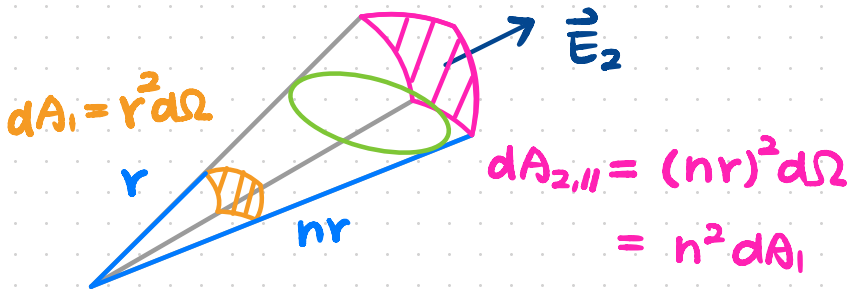
$$\Rightarrow \boxed{E = \frac{\sigma}{2\epsilon_0}} \quad \text{Infinite uniform sheet}$$

Coulomb's Law $\rightarrow$ Gauss's Law

① charge inside the surface



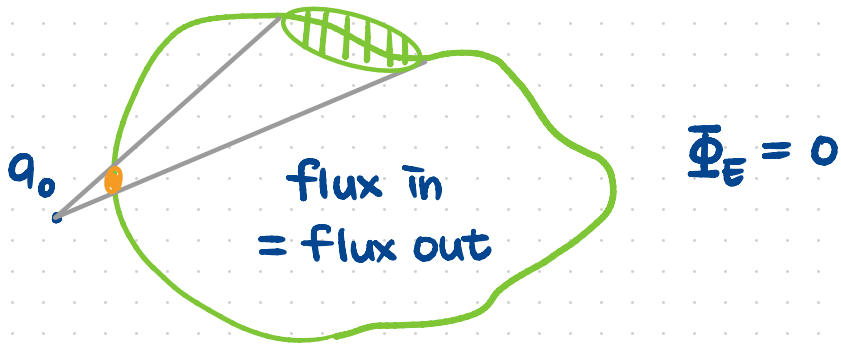
proof: flux through A_1 = flux through A_2
 $d\Phi_{E_1}$ $d\Phi_{E_2}$



$$d\Phi_{E_2} = E_2 dA_{2,\parallel} = \frac{E_1}{n^2} \times \cancel{n^2} dA_1 = E_1 dA_1 = d\Phi_{E_1}$$

$$\Phi_{E_2} = \int d\Phi_{E_2} = \int d\Phi_{E_1} = \Phi_{E_1} = \frac{q_i}{\epsilon_0}$$

② charge outside the surface



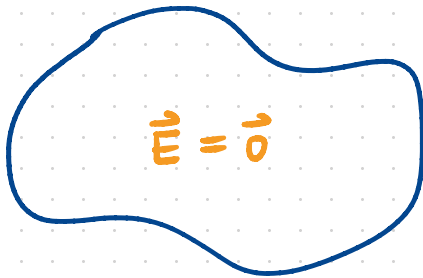
Multiple Charges

Superposition

$$\Rightarrow \Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{\sum_i q_i}{\epsilon_0} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad *$$

Properties of a conductor in electrostatic equilibrium

1. The electric field is zero inside a conductor.

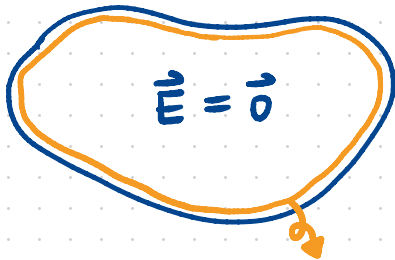


if $\vec{E} \neq \vec{0}$,

there's persistent current



2. Any excess charge resides entirely on the surface or surfaces of a conductor (by Gauss's Law)

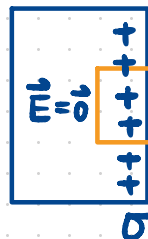
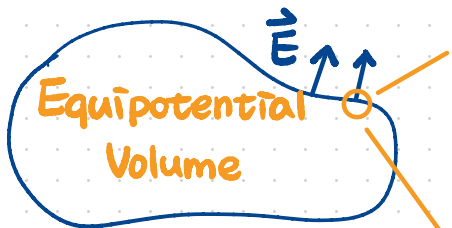


Gaussian Surface

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\Rightarrow Q_{\text{encl}} = 0$$

3. Just outside a conductor the electric field are perpendicular to its surface $\Rightarrow$ equipotential surface

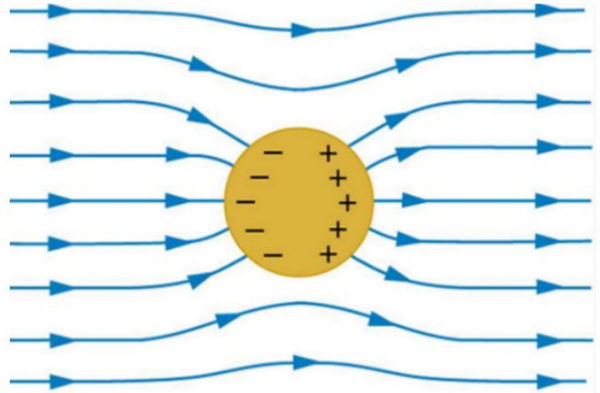
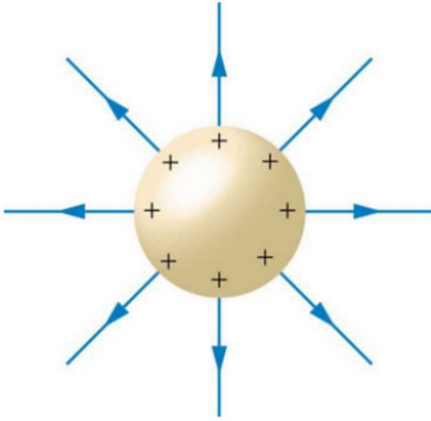


Gaussian Surface

$$\Phi_E = EA = \frac{\sigma A}{\epsilon_0}$$

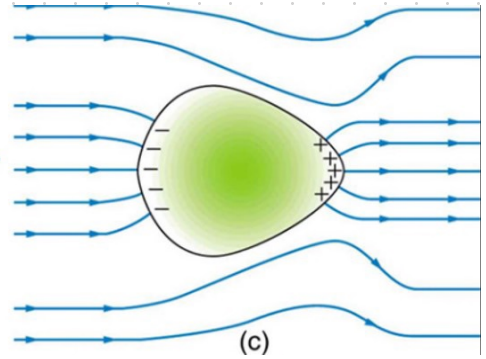
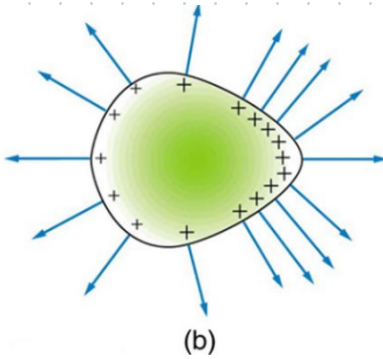
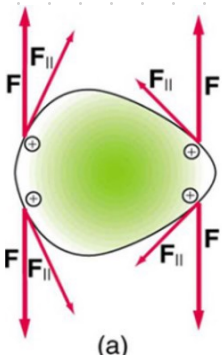
$$\Rightarrow E = \frac{\sigma}{\epsilon_0} \propto \sigma$$

A spherical conductor



<https://courses.lumenlearning.com/physics/chapter/18-7-conductors-and-electric-fields-in-static-equilibrium/>

Irregular shape



Corona Discharge 尖端放电

Air molecules become ionized, and air becomes a conductor at an electric-field magnitude of about $3 \times 10^6 \text{ V/m}$.



Copyright © 2020 Pearson Education Ltd. All Rights Reserved

Example 22.12

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

The earth (a conductor) has a net electric charge. The resulting electric field near the surface has an average value of about 150 N/C , directed toward the center of the earth.

(a) What is the corresponding surface charge density?

$$E = \frac{\sigma}{\epsilon_0}$$

$$\begin{aligned}\Rightarrow \sigma &= \epsilon_0 E = 8.85 \times 10^{-12} \times (-150) \\ &= -1.33 \times 10^{-9} \text{ C/m}^2\end{aligned}$$

(b) What is the total surface charge of the earth?

$$\begin{aligned}Q &= \sigma A = \sigma 4\pi R_E^2 \\ &= (-1.33 \times 10^{-9}) \times 4\pi \times (6.38 \times 10^6)^2 \\ &= -6.8 \times 10^5 \text{ C} \\ &= -680 \text{ kC}\end{aligned}$$

This is compensated by an equal deficiency of electron's in the earth's atmosphere.

⇒ The combination of the earth and its atmosphere is electrically neutral.

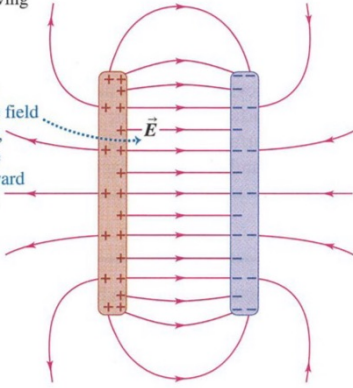
Example 22.8

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field everywhere.

(a) Realistic drawing

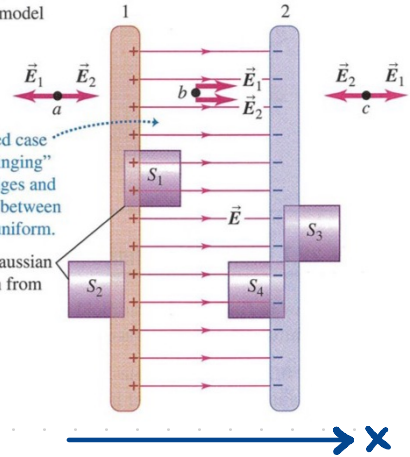
Between the two plates the electric field is nearly uniform, pointing from the positive plate toward the negative one.



(b) Idealized model

In the idealized case we ignore "fringing" at the plate edges and treat the field between the plates as uniform.

Cylindrical Gaussian surfaces (seen from the side)



$$\text{For } S_1, \vec{E} = E \hat{i}$$

$$EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{\epsilon_0}$$

$$\text{For } S_4, \vec{E} = E \hat{i}$$

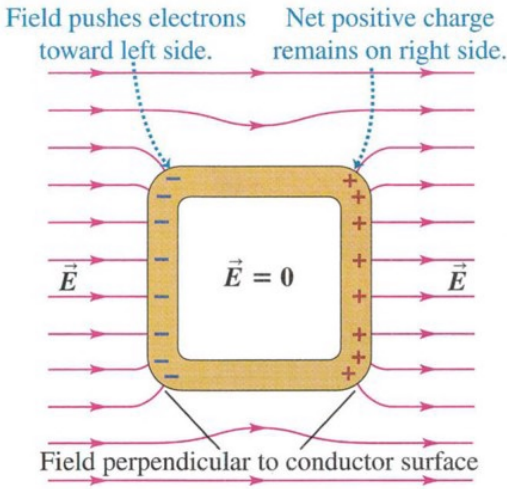
$$-EA = \frac{-\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{\epsilon_0}$$

$$\vec{E} = \vec{0} \quad \vec{E} = \frac{\sigma}{\epsilon_0} \hat{i} \quad \vec{E} = \vec{0}$$

For S_2 & S_3 ,

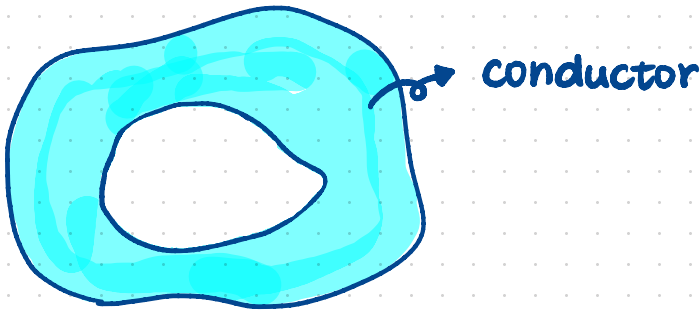
$$\vec{E} = \vec{0}$$

A Conductor with an Internal Cavity



(Courtesy Westinghouse Electric Corporation)

Faraday Cage



Electrostatic Shielding:

Charges outside a conductor have no effect on the interior of that conductor, even if the conductor has a cavity inside.

Gauss's Law

Integral Form $\Leftrightarrow$ Differential Form

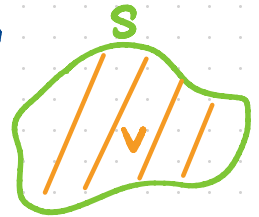
$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

LHS: Divergence Theorem

$$\oint_S \vec{E} \cdot d\vec{A} = \int_V (\vec{\nabla} \cdot \vec{E}) dV$$

RHS: $Q_{\text{encl}} = \int_V \rho dV$



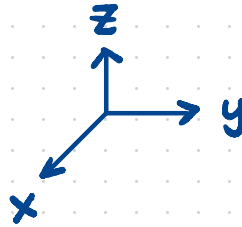
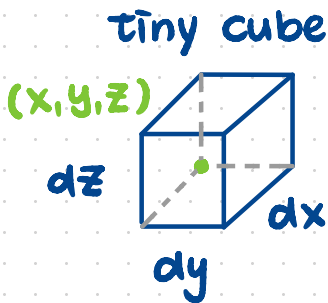
$$\Rightarrow \int_V (\vec{\nabla} \cdot \vec{E}) dV = \frac{\int_V \rho dV}{\epsilon_0}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} *$$

where $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

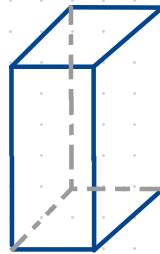
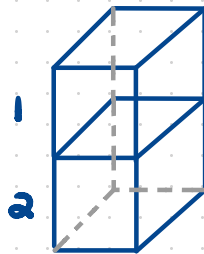
proof of Divergence Theorem



$$\begin{aligned}
 \Phi_E &= \oint \vec{E} \cdot d\vec{A} \\
 &\stackrel{\text{HW}}{=} \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) dx dy dz \\
 &= (\vec{\nabla} \cdot \vec{E}) dV
 \end{aligned}$$

Hint : $E_x(x+dx) - E_x(x) = \frac{\partial E_x}{\partial x} dx$

Then consider two cubes, and show



$$dV = dV_1 + dV_2$$

$$\oint_1 \vec{E} \cdot d\vec{A} + \oint_2 \vec{E} \cdot d\vec{A} = \oint \vec{E} \cdot d\vec{A}$$

$$= (\vec{\nabla} \cdot \vec{E}) dV_1 + (\vec{\nabla} \cdot \vec{E}) dV_2$$

$$= (\vec{\nabla} \cdot \vec{E}) dV$$

→ For an arbitrary surface



$$\sum_i \oint_i \vec{E} \cdot d\vec{A} = \sum_i \lim_{\Delta V_i \rightarrow 0} (\vec{\nabla} \cdot \vec{E}) \Delta V_i$$

$$\Rightarrow \oint_S \vec{E} \cdot d\vec{A} = \int_V (\vec{\nabla} \cdot \vec{E}) dV *$$